An Algebra of Prediction-Rewarding Mechanisms

Scoring rules, market makers, and pools as composable transducers

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Abstract

Scoring rules, market makers, parimutuels, and opinion pools compose when each stage is a stateful transducer whose message type is a distributional belief. Convex potentials generate the incentive-bearing core: proper scores from convex entropies, cost-function market makers by Fenchel conjugacy, CFMMs by convex level-set duality. The pooling and merging operators are classical: the linear and logarithmic pools are the two Kullback-Leibler barycenters, and merging market makers is the infimal convolution of risk sharing, under which liquidity adds. The benefit is illustrated by the simplest two-stage example, where stagewise equilibrium dominates conformal prediction: a residual market collects the conformal predictor's information gap $I(R;X)$ as bankroll growth while its marginal coverage stays exact.

1 Introduction

Mechanisms for eliciting and aggregating forecasts are usually studied one at a time. A proper scoring rule is analysed as a one-shot contract; a market maker as a sequential trading venue; an opinion pool as an estimator; a calibration test as a diagnostic. Real systems chain them. A forecasting platform elicits distributional predictions, aggregates them into a consensus, tests the consensus for calibration, and routes rewards back to the participants, all in a loop. The microprediction platform ran exactly this loop at scale (Cotton 2022), and the stacked-lottery design behind it was presented at MIT CSAIL in 2020 (Cotton 2020, slides 29-31): market-implied percentiles are themselves the subject of further games, and calibration is achieved by composing monotone transformations, each contributed by one or more algorithms.

Composition is also where guarantees break. Each stage of such a pipeline can be individually truthful while the chain is not. The companion paper on point-cloud scoring (Cotton 2026b) gives a concrete example: inserting an innocuous-looking kernel-density smoothing step between the submission and the score changes the optimal report from the forecaster's belief to its deconvolution.

Conformal prediction supplies a second motivation, from the opposite direction. Split-conformal is itself a composition, a point predictor chained into a rank-based calibration stage, but a degenerate one: the pool step is skipped by fiat, all credit assigned to a single model in advance, and the calibration stage prices the residual flat in the input. The wasted log score of a single-shape conformal predictor is the mutual information $I(R;X)$ between residual and input, and an entrant to a parimutuel residual pool who conditions on the input collects it as bankroll growth at exactly that rate (Cotton 2026a). Running the residual stage as an actual pool, rather than assuming its winner, is what the operators below are for; Proposition 4 locates the gap in the theory of the probability integral transform, and §5 works the example in closed form.

This note organizes the catalogue around four questions that are often blurred together:

1. *Message closure.* Can every stage consume and emit one common object, so that stages plug together at all?
2. *Convex generation.* Is each individual mechanism generated by a convex potential, so that one dictionary covers scoring rules, market makers, and their duals?
3. *Propriety under transformation.* When a stage transforms forecasts or outcomes, does strict propriety survive?
4. *Strategic composition.* When stages couple participants' payoffs, do the single-stage incentive properties survive equilibrium play?

The first is an engineering convention (§2, §6). The second and third admit theorems, which are stated with complete proofs in §3 and §4; none of the mathematics is new, but the proofs are collected in one place and one convention. The fourth is open beyond the special cases analysed in the companion paper, and §7 records what is known.

2 Preliminaries

The finite setting. Sections 3 and 4 work with a finite outcome set $\{1, \ldots, n\}$ and reports p in the probability simplex Δ. The continuous theory uses proper scoring rules on probability measures (log score, CRPS, energy and kernel scores) or discretized approximations, and the finite statements below transfer with the usual measure-theoretic care; the one place where the transfer is not routine is the probability integral transform, treated separately in Proposition 4.

Scores. A scoring rule assigns $S(p, i) \in \mathbb{R} \cup \{-\infty\}$ to a report p and outcome i; the expected score of reporting p under belief q is $S(p; q) = \sum_i q_i S(p, i)$, assumed well defined with $S(q; q)$ finite (this is the regularity used throughout). The rule is *proper* if $S(q; q) \geq S(p; q)$ for all p, q and *strictly proper* if equality forces $p = q$. Scores are written in reward form: higher is better.

Convex tools. For convex G with subgradient selection $G'(p) \in \partial G(p)$, the Bregman divergence is $D_G(q, p) = G(q) - G(p) - \langle G'(p), q - p \rangle$. The Legendre-Fenchel conjugate of a function R on Δ is $R^*(\mathbf{q}) = \sup_{p \in \Delta} \langle p, \mathbf{q} \rangle - R(p)$. The infimal convolution of f and g is $(f \,\square\, g)(x) = \inf_y f(y) + g(x - y)$. "Closed proper convex" is used in the standard sense (Rockafellar 1970). Differentiable statements are on the relative interior of Δ, with extended-real values allowed on the boundary.

3 One potential, three mechanisms

Theorem 1 (characterisation of proper scoring rules; Savage (1971), McCarthy (1956)). *A regular scoring rule S is proper iff there is a convex $G : \Delta \to \mathbb{R}$ with*

$$S(p, i) \;=\; G(p) + \langle\, G'(p),\; e_i - p \,\rangle, \qquad G'(p) \in \partial G(p),$$

where e_i is the i-th unit vector. It is strictly proper iff G is strictly convex relative to Δ, and then $G(p) = S(p; p)$ is the expected score of a truthful forecaster.

Proof. ($\Leftarrow$) Since $\sum_i q_i(e_i - p) = q - p$, the expected score is affine in the belief: $S(p;q) = G(p) + \langle G'(p), q-p\rangle$. Hence

$$S(q;q) - S(p;q) \;=\; G(q) - G(p) - \langle G'(p),\, q-p\rangle \;=\; D_G(q,p),$$

which is non-negative by the supporting-hyperplane inequality, and zero only at $q = p$ when G is strictly convex relative to Δ; so S is (strictly) proper.

($\Rightarrow$) Define $G(q) := S(q;q)$. Properness says $G(q) = \sup_p S(p;q)$, and each $q \mapsto S(p;q) = \langle S(p,\cdot), q\rangle$ is affine, so G is a pointwise supremum of affine functions, hence convex. For any fixed p,

$$S(p;q) \;=\; G(p) + \langle S(p,\cdot),\, q-p\rangle \quad \text{with} \quad S(p;q) \le G(q) \;\; \forall q,$$

so the vector $S(p,\cdot)$ is a subgradient of G at p (acting on the tangent space of Δ, where subgradients are defined up to a constant shift along $\mathbf{1}$, which the representation absorbs). Substituting $G'(p) = S(p,\cdot)$ into the display returns $S(p,i)$ identically. If S is strictly proper the supremum has a unique maximiser at every q, which for a supremum of affine functions is equivalent to strict convexity relative to Δ. ■

The content of Theorem 1 is the dictionary *proper scoring rule* $\leftrightarrow$ *convex function* $\leftrightarrow$ *Bregman divergence* (Gneiting and Raftery 2007; Banerjee et al. 2005). The classics are three choices of G, with the divergences computed directly from the definition:

Score	Generator $G(p) = S(p;p)$	Bregman divergence $D_G(q,p)$
logarithmic	$\sum_i p_i \log p_i$	$\mathrm{KL}(q\\|p)$
Brier (quadratic)	$\\|p\\|_2^2$	$\\|q-p\\|_2^2$
spherical	$\\|p\\|_2$	$\\|q\\|_2\,(1-\cos\theta_{p,q})$

For the spherical row: $\nabla G(p) = p/\|p\|$, so $D_G(q,p) = \|q\| - \langle p,q\rangle/\|p\| = \|q\|(1-\cos\theta_{p,q})$, an angular term scaled by $\|q\|$. For the logarithmic row the representation gives $S(p,i) = \log p_i$ on the relative interior, with $-\infty$ on the boundary.

Theorem 2 (scoring rule to market maker; Hanson (2007), Abernethy et al. (2013)). *Let R be closed, proper, and strictly convex on Δ (in the sequel, $R = G$, the generator of Theorem 1 read as a regulariser), and define*

$$C(\mathbf{q}) = \sup_{p\in\Delta} \left(\langle p, \mathbf{q}\rangle - R(p)\right).$$

Then:

(i) C is convex and finite, the maximiser $p^*(\mathbf{q})$ is unique, and $\nabla C(\mathbf{q}) = p^*(\mathbf{q}) \in \Delta$: prices are a probability vector. (Without strict convexity, prices live in $\partial C(\mathbf{q})$.)

(ii) C is translation-equivariant, $C(\mathbf{q} + \alpha\mathbf{1}) = C(\mathbf{q}) + \alpha$; costs telescope over any trade path, so round trips cost zero and buying the full bundle costs its payout: the market is arbitrage-free.

(iii) With initial state $\mathbf{q}_0 = \mathbf{0}$, the maker's loss when the market settles on outcome i after net sales $\mathbf{q}$ is

$$\mathrm{loss}_i(\mathbf{q}) = q_i - C(\mathbf{q}) + C(\mathbf{0}) \;\le\; R(e_i) - \inf_{p\in\Delta} R(p) \;\le\; \sup_{p\in\Delta} R(p) - \inf_{p\in\Delta} R(p).$$

(iv) Taking $R(p) = b\sum_i p_i \log p_i$ gives $C(\mathbf{q}) = b\log\sum_i e^{q_i/b}$, Hanson's LMSR, with worst-case loss $b\log n$.

Proof. (i) C is a supremum of affine functions of $\mathbf{q}$, hence convex; the supremum of a continuous function over the compact Δ is attained, and strict concavity of $p \mapsto \langle p, \mathbf{q}\rangle - R(p)$ makes the maximiser unique. Danskin's theorem gives $\nabla C(\mathbf{q}) = p^*(\mathbf{q})$. (ii) $\langle p, \mathbf{q} + \alpha\mathbf{1}\rangle = \langle p, \mathbf{q}\rangle + \alpha$ for $p \in \Delta$, so the supremum shifts by α. A trader moving the state $\mathbf{q} \to \mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$ by definition, so costs over any path telescope; a round trip costs zero, and by translation equivariance the bundle $\alpha\mathbf{1}$ costs exactly α, its sure payout. (iii) The maker collects $C(\mathbf{q}) - C(\mathbf{0})$ and pays q_i, so $\text{loss}_i = q_i - C(\mathbf{q}) + C(\mathbf{0})$. By Fenchel-Moreau, $\sup_{\mathbf{q}} (\langle e_i, \mathbf{q}\rangle - C(\mathbf{q})) = R^{**}(e_i) = R(e_i)$, and $C(\mathbf{0}) = \sup_p -R(p) = -\inf_p R(p)$. The final inequality holds because $e_i \in \Delta$; for convex R the supremum over Δ is attained at an extreme point, so it is typically an equality. (iv) Lagrange: maximising $\langle p, \mathbf{q}\rangle - b\sum p_i \log p_i$ subject to $\sum p_i = 1$ gives $q_i - b(\log p_i + 1) = \lambda$, so $p_i \propto e^{q_i/b}$, and substituting back yields $C(\mathbf{q}) = b\log\sum_i e^{q_i/b}$. Then $R(e_i) = 0$ and $\inf R = -b\log n$ at the uniform distribution, so the loss bound is $b\log n$. ■

Proposition 3 (cost-function markets and CFMMs). *Under the monotonicity, concavity, and reserve-domain hypotheses of Frongillo et al. (2024), cost-function prediction markets and constant-function market makers can be converted into one another. In the unconstrained-reserve sign convention a cost function C gives a concave CFMM potential by $\varphi(\mathbf{r}) = -C(-\mathbf{r})$; bounded-reserve versions require a perspective (level-set) construction. The equivalence is a convex level-set duality; it is not the bare Fenchel conjugacy $C \mapsto C^*$.*

We do not reprove the general equivalence here; Frongillo et al. (2024) give it in full, and Angeris et al. (2023) and Angeris et al. (2021) develop the CFMM side.

Example (constant product). Take two outcome tokens with reserves r_1, r_2 and invariant $r_1 r_2 = k$. The pool's portfolio value at prices $(p, 1-p)$ is

$$V(p) = \inf\{p\, r_1 + (1-p)\, r_2 : \; r_1 r_2 = k\} = 2\sqrt{k\, p(1-p)},$$

by the AM-GM inequality, with the infimum attained at $r_1 = \sqrt{k(1-p)/p}$. V is concave; reading $R(p) = -V(p)$ as the regulariser of Theorem 2 gives a maker whose worst-case loss is the range of R on $[0,1]$, namely $\sqrt{k}$ (between $p = \frac{1}{2}$ and the endpoints): the geometric mean of the initial reserves. The corresponding generator is not entropic, which is the convex-analytic content of the observation that constant-product markets and the LMSR price bounded-payout claims differently.

Proposition 4 (the probability integral transform; Rosenblatt (1952), Dawid (1984)). *If X has continuous CDF F then $U = F(X) \sim \mathrm{Uniform}(0,1)$, and $z = \Phi^{-1}(U) \sim N(0,1)$. In the prequential setting, if F_t is the true conditional law of X_t given the past, then $U_t = F_t(X_t)$ is conditionally uniform and the PIT stream is iid uniform.*

Proof. For continuous F, with the generalized inverse $F^-(u) = \inf\{x : F(x) \geq u\}$, the event $\{F(X) \leq u\}$ differs from $\{X \leq F^-(u)\}$ by a set of probability zero, and $\Pr(X \leq F^-(u)) = F(F^-(u)) = u$ by continuity. The prequential statement applies this conditionally at each t. ■

Two scope remarks. First, the converse fails in the strong sense: marginally uniform PITs do not certify an informative forecast. Reporting the unconditional law F of an iid sequence gives $F(X_t) \sim \mathrm{Uniform}(0,1)$ even if valuable covariates were ignored. A PIT critic therefore witnesses miscalibration relative to its test class; it complements, and does not replace, a proper score that rewards sharp conditional distributions. Conformal prediction lives on exactly this distinction: a split-conformal predictor achieves marginal coverage by construction, the uniform-PIT guarantee, while free to ignore the conditional information in the input. For a single-shape conformal predictor the wasted log score equals the mutual information $I(R;X)$ between residual and input, and an entrant to the parimutuel pool of the companion paper who conditions on X collects it as bankroll growth at exactly that rate; the flat-in-X conformal report is the break-even crowd (Cotton 2026a). Second, for discrete forecasts the randomized PIT preserves the exact uniform null, while the mid-PIT is a convenient deterministic diagnostic with a different null distribution.

4 Operators on mechanisms

The common signature. A *transducer* (Mealy machine) is a tuple $(S, A, B, \delta, \lambda)$ with state space S, input alphabet A, output alphabet B, transition map $\delta : S \times A \to S$, and output map $\lambda : S \times A \to B$; run on an input stream it produces $s_{t+1} = \delta(s_t, a_t)$ and $b_t = \lambda(s_t, a_t)$, a causal map of input streams to output streams. The mechanisms of §3 share one instantiation. Let Dist denote the set of distributional beliefs and take S the wealth states, $A = \mathrm{Dist}^m \times \mathcal{X}$ (m participant reports and, where the stage settles, a realized outcome), and $B = \mathrm{Dist} \times \mathbb{R}^m$ (an aggregate belief and transfers). A *stage* is such a transducer, written

$$M : (\mathrm{Dist}^m,\ w,\ x) \longmapsto (\mathrm{Dist},\ w',\ \pi).$$

A scoring rule is the transfer component λ of a stage, not itself a map Dist $\to$ Dist; a market maker is a stage whose state is the inventory vector and whose emitted belief is the price; an opinion pool is a stage with no outcome argument and zero transfers. Composition wires the belief output of one stage to the belief inputs of the next while state and transfers thread through.

The operators below act on stages.

Sequentialise. Theorem 2: a proper score run sequentially against a wealth state is a cost-function market maker.

Pool. A proper score gives a batch elicitation mechanism when reports are scored independently and funded externally: properness is inherited report by report. Parimutuel and budget-balanced versions are a different game, because the pot split couples payoffs through the denominator; the companion paper's §2 gives the price-taking analysis (truthful all-in, a symmetric equilibrium at fractional stakes, degenerate as the stake fraction vanishes), and beyond that the equilibrium theory is open.

Ensemble (Proposition 5: the two pools are the two KL barycenters). *Let $q_1, \ldots, q_m$ be densities and $w_i \geq 0$, $\sum w_i = 1$. Then the linear pool minimizes the forward divergences and the logarithmic pool the reverse:*

$$\arg\min_p \textstyle\sum_i w_i \,\mathrm{KL}(q_i \| p) = \sum_i w_i q_i, \qquad \arg\min_p \sum_i w_i \,\mathrm{KL}(p \| q_i) \;\propto\; \prod_i q_i^{w_i}.$$

Proof. For the first, $\sum_i w_i \,\mathrm{KL}(q_i \| p) = \mathrm{const} - \int (\sum_i w_i q_i) \log p$, and $\int \bar{q} \log p$ is maximized over densities at $p = \bar{q}$ (Gibbs). For the second, $\sum_i w_i \,\mathrm{KL}(p \| q_i) = \int p \log p - \int p\, \overline{\log q}$ with $\overline{\log q} = \sum_i w_i \log q_i$; the Lagrange condition is $\log p = \overline{\log q} + \mathrm{const}$. ■

Training weights by cumulative log score and then mixing linearly, $\sum_m w_m F_m$, is Bayesian model averaging, a linear pool with score-trained weights; the logarithmic pool multiplies densities and renormalizes. They are different aggregates with different sharpness (Genest and Zidek 1986).

Merge (Proposition 6: merging makers is infimal convolution; Rockafellar (1970), Bhaskara et al. (2023)). *For closed proper convex f, g: $(f \mathbin{\square} g)^* = f^* + g^*$. Consequently, merging two cost-function makers with regularisers R_1, R_2 (cost functions $C_i = R_i^*$) yields the maker with regulariser $R_1 + R_2$, and merging LMSR_{b_1} with LMSR_{b_2} yields $\mathrm{LMSR}_{b_1+b_2}$: liquidity adds.*

Proof. $(f \mathbin{\square} g)^*(p) = \sup_x \langle p, x\rangle - \inf_y\{f(y) + g(x-y)\} = \sup_{y,z} \langle p, y\rangle - f(y) + \langle p, z\rangle - g(z) = f^*(p) + g^*(p)$. For the makers, $C_1 \mathbin{\square} C_2 = (R_1 + R_2)^*$ by biconjugacy, and $b_1 G + b_2 G = (b_1 + b_2)G$ for the entropic generator. ■

This is also the risk-sharing literature's composition law: the aggregate of several agents' convex risk measures is their infimal convolution, with the Pareto allocation as minimiser and the common subgradient as the clearing price (Barrieu and El Karoui 2005; Jouini et al. 2008); the market-liquidity reading is developed by Bhaskara et al. (2023).

Conjugate. Run a stage in a transformed coordinate and map back. This is where propriety needs care, and the theory splits in two:

- *Fixed transformations.* Properness is preserved under any fixed transformation of forecast and outcome, and strict propriety iff the transformation is injective (Allen et al. 2023, Prop. 4; Pic et al. 2025, Prop. 1). The Markov-kernel (stochastic channel) extension, strict propriety iff the channel is injective on laws, is Theorem 2 of the companion point-cloud paper, whose Theorem 1 also exhibits the canonical failure: a KDE smoothing seam whose raw-outcome score elicits a deconvolution.
- *Forecast-dependent transformations.* The PIT critic transforms outcomes by the reported F itself, so the fixed-transformation theorems do not apply to it; Proposition 4 and its scope remarks are the correct warrant, and the critic is a calibration diagnostic rather than a strictly proper score. The stacked-lottery design (Cotton 2020, slides 29-31) is this operator in practice: percentiles from one game feed the next, and calibration is produced by composing monotone maps contributed by competing algorithms.

Residual. Let a stage emit the aggregate F_1, and let a second market elicit a distribution for the residual $U = F_1(X)$, settling at the realised $u = F_1(x)$. If F_1 is the true conditional law then U is uniform (Proposition 4) and the second market has nothing to price; whatever structure remains in the residual is the second stage's edge. The corrected forecast composes the two reports.

Proposition 7 (the correction is a multiplicative reweighting). *Let F_1 be strictly increasing with density $p_1 > 0$ and let the residual market's consensus be a distribution G on $[0, 1]$ with density g. The composed forecast $F = G \circ F_1$ has density*

$$p(x) \;=\; p_1(x)\, g\big(F_1(x)\big),$$

so $\log p(x) = \log p_1(x) + \log g(u)$ with $u = F_1(x)$: the chain's log score is the sum of stage log scores, and the residual stage is paid by a proper score on u alone.

Proof. Chain rule: $F'(x) = g(F_1(x))\, p_1(x)$; take logarithms. The residual score $\log g(u)$ is the logarithmic score of Theorem 1 applied to the report g and outcome u, hence strictly proper for the law of U. ■

Multiplying the density by a ratio fitted to what the current model gets wrong is the functional-gradient step of boosting under log loss (Mason et al. 1999; Friedman 2001), so a chain of residual markets is stagewise boosting with wealth as the learning rate. What Proposition 7 does not settle is the game across stages: who funds the residual pot, and whether a forecaster free to enter both stages prefers to withhold information from the first and sell it to the second (§7).

Spec. Serialise a pipeline to data and search over it; the mechanism analogue is a market over pipelines. Also open.

Stagewise play. Call a profile of reports a *stagewise equilibrium* if no participant gains by a deviation confined to a single stage, all other stages' reports held fixed. This is the pipeline version of the myopic-trader assumption standard in the market-scoring-rule literature (Hanson 2007; Chen and Vaughan 2010).

Proposition 8 (single-stage guarantees compose under stagewise play). *Suppose each stage of a pipeline, taken in isolation with its inputs fixed, makes the truthful report a best response: strict propriety for externally funded scoring stages (Theorem 1), the sequential scoring of Theorem 2 for market stages, the price-taking pot-split analysis of the companion paper for parimutuel stages, and the residual score of Proposition 7 for correction stages. Then truthful reporting at every stage is a stagewise equilibrium of the pipeline.*

Proof. A deviation confined to stage k changes the deviator's payoff only through stage k's transfer map, since every other stage's reports, and hence its inputs, are held fixed; the stage-k hypothesis then makes the truthful report a best response. ■

What the concept excludes is cross-stage deviation, the derivative-on-the- underlying play of §7.

5 Worked composition: pricing the conformal residual

The introduction claimed that split-conformal prediction is a degenerate composition and that running the residual stage as a market collects what the degeneracy wastes. This section proves it in the simplest setting that carries the full structure.

Let the residual left by a fixed point predictor remain correlated with the input: (R, X) jointly Gaussian with correlation ρ, marginally $R \sim N(0, \sigma_R^2)$ and $X \sim N(0, \sigma_X^2)$. Any misspecified linear predictor produces this. The split-conformal wrapper reports the marginal law of R at every x; its coverage is exact at every level, since the marginal PIT $F_R(R)$ is uniform (Proposition 4). The wrapper passes the only test it runs.

Now run the residual stage as a market of the Residual operator: reports are densities for R, settled by the logarithmic score. The wrapper's report is $N(0, \sigma_R^2)$, flat in x; an entrant who conditions reports the conditional law $R \mid X = x \sim N(\rho(\sigma_R/\sigma_X)\, x,\ (1-\rho^2)\sigma_R^2)$.

Proposition 9 (the rent). *Per observation, the conditioning entrant's expected log-score edge over the flat report is*

$$\mathbb{E}\left[\mathrm{KL}(\, p_{R|X} \,\|\, p_R\,)\right] \;=\; -\tfrac{1}{2}\log(1-\rho^2) \;=\; I(R;X),$$

a Kelly-staked entrant's bankroll grows at exactly this rate against the flat crowd, and the composed forecast of Proposition 7 built from the entrant's report recovers the conditional law exactly.

Proof. With $m(x) = \rho(\sigma_R/\sigma_X)x$ and $s^2 = (1-\rho^2)\sigma_R^2$,

$$\mathrm{KL}(N(m, s^2) \,\|\, N(0, \sigma_R^2)) = \log\frac{\sigma_R}{s} + \frac{s^2 + m^2}{2\sigma_R^2} - \frac{1}{2},$$

and $\mathbb{E}[m(X)^2] = \rho^2\sigma_R^2$, so the expectation is $-\frac{1}{2}\log(1-\rho^2)$, the mutual information of a bivariate Gaussian pair. The Kelly growth rate is $\mathbb{E}[\log p_{R|X}(R) - \log p_R(R)]$, the same expectation. For the recovery, the entrant's report on the residual rank $U = F_R(R)$ has conditional density $g(u \mid x) = p_{R|X}(F_R^-(u))/p_R(F_R^-(u))$, and Proposition 7 composes to $p_R(r)\, g(F_R(r) \mid x) = p_{R|X}(r)$. ■

At $\rho = \frac{1}{2}$ the rent is 0.144 nats per observation and the entrant's bankroll doubles every five observations; at $\rho = 0.3$, every fifteen. Throughout, the conformal band's marginal coverage remains exact: the wrapper's own diagnostic is uniform while the transfer runs, which is the sense in which marginal coverage is the wrong audit for a composed system. The wealth flow is the mechanism's estimator of $I(R; X)$, the sequential counterpart of the static bounds in (Cotton 2026a).

6 Implementation notes

The mechanisms of the catalogue are implemented against a single interface patterned on the skaters time-series contract (Cotton 2026c), a successor to timemachines (Cotton 2021): every stage consumes and emits the same distributional type and threads its own state, so the signature of §4 is the forecasting contract with wealth in place of model state. The closure of the message type is what makes a small operator set sufficient; the transforms, ensembles, and residual constructions of §4 all consume and emit the one type. Scores are implemented in loss form; the text uses reward form.

7 Open problems

1. *Strategic composition.* Proposition 8 settles stagewise play; deviations that span stages are open. A downstream stake is a derivative written on an upstream settlement, and the derivative position creates an incentive to distort the underlying. Cash-settled futures reward manipulating the settlement price (Kumar and Seppi 1992), option positions reward moving the underlying (Jarrow 1994), and in prediction markets anticipated manipulation is priced in and can even sharpen the aggregate (Hanson and Oprea 2009; Ostrovsky 2012). Which of these results transfer across mechanism seams is the question.
2. *Conservation of edge.* Wealth conservation is automatic, a sum of stage-level zeros, and is not the invariant. The invariant that can fail is the edge a truthful participant holds over the price, $D_G(q, \pi)$. For the logarithmic score the edge is $\mathrm{KL}(q\|\pi)$ and the question is closed by the data-processing inequality: an interface channel cannot increase it, and preserves it exactly when the channel is sufficient for the pair (Csiszár 1967). Open is the same statement for the Bregman edge of a general generator G, and the rate at which edge leaks through an approximately non-sufficient interface.
3. *Residual markets.* Proposition 7 gives the single-step correction; a chain of residual markets is stagewise boosting with wealth as the learning rate. Open: the microstructure (who funds each residual pot, and when it settles relative to the stage before), whether a forecaster free to enter several stages prefers withholding information upstream to sell it downstream, and whether the iterated correction converges to the true conditional law as boosting does. Split-conformal prediction is the one-participant degenerate case, with its rent $I(R; X)$ as the value left on the table (Cotton 2026a).

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